

Status. THM052 has passed both complete fresh audit axes and root comparison. The original critical source assertion remains open.

Fix $N \geq 2$ and $0 < \nu < \infty$. Let $\Omega \subset (\mathbb{T}^4)^N$ be the collision-free configuration space, with unit product Haar measure dx . For the coefficient-one periodic Coulomb kernel,

$$\widehat{g}(m) = |m|^{-2} \ (m \neq 0), \quad \widehat{g}(0) = 0, \quad H(x) = \frac{1}{N} \sum_{i < j} g(x_i - x_j), \quad c = 4\pi^2, \quad \kappa = c(N-1).$$

On Ω , $\Delta H = \kappa$. The full distributional identity retains the collision atom; it is not used as a pointwise identity there. The original process and an explicitly auxiliary process solve

$$dX = -\nabla H(X)dt + \sqrt{2\nu} dW, \quad dY = +\nabla H(Y)dt + \sqrt{2\nu} dW.$$

The original process is conservative. The auxiliary process is killed at its maximal lifetime ζ in Ω , with no entrance rule at collision. Define

$$P_t h(x) = \mathbb{E}_x h(X_t), \quad Q_t f(x) = \mathbb{E}_x [f(Y_t) \mathbf{1}_{\{t < \zeta\}}].$$

Accepted statement. For bounded nonnegative measurable f, h , every $t \geq 0$ satisfies

$$\int f P_t h \, dx = e^{\kappa t} \int h Q_t f \, dx. \tag{1}$$

The identity extends to bounded signed or complex tests by linearity. From iid Haar, the actual repulsive density at time t is $e^{\kappa t} Q_t 1$, hence bounded by $e^{\kappa t}$. In the auxiliary attractive process,

$$\mathbb{E}_{\text{Haar}} [f(Y_t) \mathbf{1}_{\{t < \zeta\}}] = e^{-\kappa t} \int f \, dx, \quad \mathbb{P}_{\text{Haar}}(\zeta > t) = e^{-\kappa t}. \tag{2}$$

Thus this lifetime has an exact exponential distribution, and its current configuration conditional on survival is product Haar. These statements are for the specified initial law; the point-start survival probability need not be constant.

For each finite $T \geq 0$, the complete original path law equals

$$\mathcal{L}_{\text{Haar}}((X_t)_{0 \leq t \leq T}) = \mathcal{L}_{\text{Haar}}((Y_{T-t})_{0 \leq t \leq T} \mid \zeta > T), \tag{3}$$

as probability measures on $C([0, T], (\mathbb{T}^4)^N)$ with the uniform topology. This does not make the original positive-time law product Haar.

Construction and finite-domain identity. Locally $g(z) = |z|^{-2}$ plus a smooth even remainder. Therefore H is bounded below and diverges at every partial collision. Local Lipschitz equations construct both paths up to compact exits, with measurable, unique restart. For the original path, stopped Itô calculus gives

$$dH(X_t) = \{-|\nabla H(X_t)|^2 + \nu\kappa\}dt + \sqrt{2\nu}\nabla H(X_t) \cdot dW_t.$$

The shifted energy is a nonnegative barrier; its expected stopped value bounds the probability of reaching height R by $C_{x,\nu,T}/R$. Letting R increase proves noncollision and conservativity. No such conservativity is assumed for the auxiliary process.

Use increasing domains $D_L = \{\min_{i<j} \text{dist}(x_i, x_j) > 1/L\}$, whose closures lie in Ω . Extend the two drifts smoothly and boundedly near each closure, and kill upon exiting D_L . Boundary corners cause no difficulty in the path argument below. All following bounds are at fixed L, N, ν .

Write $D = D_L$ and let B be torus Brownian motion with generator $\nu\Delta$. On paths remaining in D ,

$$\int_0^t \nabla H(B_s) \cdot dB_s = H(B_t) - H(B_0) - \nu\kappa t.$$

For a bounded drift b , the Girsanov density is

$$\exp \left\{ \frac{1}{2\nu} \int_0^t b(B_s) \cdot dB_s - \frac{1}{4\nu} \int_0^t |b(B_s)|^2 ds \right\}.$$

Its martingale condition holds because the drift extension is bounded. Set

$$V = \frac{|\nabla H|^2}{4\nu}, \quad K_t^D v(x) = \mathbb{E}_x^B \left[\mathbf{1}_{\{\tau_D > t\}} e^{-\int_0^t V(B_s) ds} v(B_t) \right].$$

Stationary torus Brownian motion is reversible by its symmetric heat densities. The event of staying in D and the time integral of V are invariant under reversal, so K_t^D is Haar symmetric. Bounded-domain Girsanov then gives

$$P_t^D h = e^{H/(2\nu) + \kappa t/2} K_t^D (e^{-H/(2\nu)} h), \quad Q_t^D f = e^{-H/(2\nu) - \kappa t/2} K_t^D (e^{H/(2\nu)} f).$$

All multipliers are bounded on this fixed domain. Symmetry gives (1) for the killed-domain semigroups. Increasing-domain killed expectations are monotone for nonnegative tests. Their limits are respectively the conservative original semigroup and the minimal attractive semigroup killed at ζ . Monotone convergence gives (1) on Ω , with no formal collision boundary integration or uniform exponential-energy estimate. Setting $f = 1$ gives the density; setting $h = 1$ gives (2).

Full path identification. For $0 = t_0 < \dots < t_m = T$ and bounded nonnegative h_j , repeated transposition by (1) yields

$$\begin{aligned} & \int h_0 P_{t_1-t_0} (h_1 P_{t_2-t_1} (\dots h_m)) dx \\ &= e^{\kappa T} \int h_m Q_{t_m-t_{m-1}} (h_{m-1} Q_{t_{m-1}-t_{m-2}} (\dots h_0)) dx. \end{aligned}$$

The right side is exactly the attractive reversed cylinder expectation with survival indicator, divided by its survival probability. On survival beyond T the path is continuous. Rational-time evaluations generate the Borel sigma-field of the uniform continuous-path space, so equality of these cylinders proves (3) as an equality of full path laws. No additional tightness argument or endpoint-only inference is used; $T = 0$ is the initial-law identity.

The independent reconstruction also proves (3) directly: at a fixed continuous driving path, the repulsive endpoint flow has Jacobian $e^{-\kappa T}$, and its inverse is the attractive flow driven by reversed increments on the corresponding surviving sets. Change of variables and Wiener reversal give the same full path identity.

Exact defect and the unresolved limit. For every bounded complex configuration observable F and $A \in \mathbb{C}$, (3) gives

$$\mathbb{E}_{\text{rep}} |F(X_T) - AF(X_0)|^2 = \mathbb{E}_{\text{att}} [|F(Y_0) - AF(Y_T)|^2 \mid \zeta > T]. \quad (4)$$

Bounded measurable F also follows directly from the endpoint kernel identity. For fixed nonzero k , take

$$F(x) = \frac{1}{N} \sum_i e^{2\pi i k \cdot x_i}, \quad A = e^{-(c+\nu c|k|^2)T}.$$

Multiplying (4) by N gives the identical modal endpoint defect used in THM049. Along $\beta_N/\sqrt{N} \rightarrow \lambda \in (0, \infty)$, with $\nu_N = \beta_N^{-1}$, its vanishing and positive-limsup alternatives therefore agree exactly with those of the conditional attractive expression. The conditioning probability is $e^{-c(N-1)T}$; it cannot be dropped from an estimate. No decay rate, positive-limsup witness, endpoint independence, original product law, or new fluctuation theorem follows from this representation alone.